CG2 Glaze Search · System Architecture (Information Flow)

FLOW 1 · BUILD THE DATABASE

Runs once per data update: turns recipe photos into reviewed, described ceramic objects

1 DATA SOURCES

- **Glazy recipes**
  7k CSV · ingredients
- **7,063 photos**
  + 125 bonus photos

2 FIND OBJECTS

- **Grounding DINO**
  text prompt → boxes
- **SAM 2.1**
  box → exact mask
- **CLIP**
  is it ceramic?

3 DESCRIBE

- **Colour regions**
  3 main colours · ΔE2000
- **Six attributes**
  top-3 candidates
- **Look + shape**
  DINOv2 · outline

4 REVIEW

- **Review app**
  local + online
- **Corrections**
  390 + 251 objects
- **Not needed**
  3,010 objects

5 INDEX

- **Embeddings**
  look · shape · surface
- **Search gallery**
  12,309 objects

6 SHARED STORAGE

- **Object crops + masks**
- **Review CSV + Tile**
- **Search index**

FLOW 2 · SEARCH

On demand: answers a user's photo, words or formula

A YOUR PHOTO

- **Drop · paste · link**
- **Click the object**

B SAME PIPELINE

- **DINO + SAM + CLIP**
- **Colours + attributes**

C COMPARE + WEIGH

- **Shape · glaze · colour**
  priorities add to 1
- **Score = weighted sum**

D RESULTS

- **Ranked objects**
  + their recipes
- **Evaluation**
  Hit · Precision · Recall

FINAL PURPOSE

Glaze photos → reviewed, described ceramic objects → the right recipe found by look, shape and colour

How an image is processed

FLOW 1 · IMAGE → OBJECTS → DESCRIPTION

The same steps run for the database photos and for every uploaded photo

1 PHOTO

- **Original photo**
  never changed
- **Recipe link**
  photo id = Glazy id

2 GROUNDING DINO

- **Text prompt**
  ceramic object · pottery piece · ceramic test tile · ceramic test sample · ceramic fragment · pottery shard …
- **Boxes + scores**
  low threshold, recall first
- **Hands excluded**
  "human hand. finger."

3 SAM 2.1

- **3 masks per box**
  + predicted quality
- **One object or many?**
  dominant vs. group
- **Split into objects**
  <id>_1, _2 …

4 CLIP CHECK

- **Ceramic score**
  ≥ 0.35 kept
- **Doubtful**
  flagged for review

5 DESCRIBE

- **Colour regions**
  ColorThief · ΔE2000
- **Six attributes**
  DINO phrases → layer
- **Glaze look**
  DINOv2 features
- **Shape**
  outline geometry

i SIX ATTRIBUTES

object form · surface finish · transparency · texture · glaze effects · defects

- **Top-3 candidates**
  probability + where found

ii TRAINED ON

- **Claude's visual labels**
  1,504 train · 481 test
- **Right first / in top 3**
  form 84 % / 92 %

iii STORED

- **Crops · masks · tags**
  per object
- **Index**
  look · shape · surface · colour

FINAL PURPOSE

Every photo becomes separate ceramic objects, each with colours, six attributes, a look and a shape

How the objects are reviewed

FLOW 3 · REVIEW *People check the machine's work; their decisions are kept apart from the machine labels*

1 OPEN

- **Review app** — OPEN_REVIEW_APP.bat
- **Online page** — editors sign in with a code

2 MARK

- **Not needed** — N key · moved aside
- **Needs review** — R key + note
- **Ignore the rest** — I key

3 CORRECT

- **Correction recipe** — per photo, from the note
- **SAM 2.1 again** — box · points · merge · split
- **Round 1 · round 2** — 390 · 251 objects

4 TILE

- **Six attributes** — confirm · reject · correct
- **Top-3 candidates** — click → box shown
- **review_tile** — T key · category

5 SYNC

- **Online → PC** — SYNC_ONLINE_REVIEWS
- **Kept with history** — CSV · tile_review.json

i STATUS

unreviewed · not needed · needs review · completed · completed v2

ii CLAUDE'S ROLE

- **Wrote correction recipes** — from your notes
- **Labelled attributes** — 1,985 objects
- **Not at search time** — no Claude call online

iii NEVER OVERWRITTEN

- **Your marks** — newer one wins
- **Machine labels** — separate files

FINAL PURPOSE

Human decisions correct the segmentation and the attributes, and are kept with their history

How objects are retrieved · criteria and their balance

FLOW 2 · SEARCH BY CATEGORY

Each database object gets one similarity per component; the priorities decide how much each counts

1 SHAPE

- **Outline geometry**
  + silhouette
- **Object form**
  bowl · cup · tile …

2 GLAZE / SURFACE

- **Glaze look**
  DINOv2 appearance
- **Surface + effects**
  attribute probabilities

3 COLOUR

- **Colour regions**
  ΔE2000 · position
- **Typed / picked**
  words · HEX

4 COMBINE

- **Priorities add to 1**
  presets or sliders
- **Weighted sum**
  of standardised scores

5 PRESETS · SHAPE / GLAZE / COLOUR

shape · glaze / surface · colour

	shape	glaze / surface	colour	
Balanced	0.20	0.45	0.35	0.20 / 0.45 / 0.35
Shape	0.65	0.20	0.15	0.65 / 0.20 / 0.15
Glaze / surface	0.05	0.85	0.10	0.05 / 0.85 / 0.10
Colour	0.05	0.15	0.80	0.05 / 0.15 / 0.80

Moving one slider shrinks the other two in proportion · the total stays 1

6 FINAL SCORE

- **Photo part**
  0.40 (ticked attributes: 40 % of it)
- **Text**
  0.35 · only when typed
- **Formula**
  0.25 · oxides + ingredients
- **Contribution**
  weight × how far above average

FINAL PURPOSE

Search by colour, shape or glaze alone, or any mix of them with a chosen balance

Evaluation · Hit@K, Precision@K, Recall@K (K = 3, 5, 10, 15, 20)

ONE REAL SEARCH *Balanced · priority shape 0.20 / glaze 0.45 / colour 0.35 · strict = shape AND glaze AND colour*



H HIT@K

Any relevant in top K?
yes / no for one search

Hit Rate@K
share of searches with a hit

P PRECISION@K

Relevant in top K ÷ K
how clean the list is

R RECALL@K

Relevant in top K ÷ all relevant
tiny when thousands match

True recipe Recall@K
its objects in the top K

Criterion · rule	K = 3	K = 5	K = 10	K = 15	K = 20
Shape same object form: cup · 784 match	✓ **100%** R 0.38%	✓ **100%** R 0.64%	✓ **100%** R 1.28%	✓ **100%** R 1.91%	✓ **100%** R 2.55%
Glaze / surface same surface group: satin (your object: satin) · 4,359 match	✓ **100%** R 0.07%	✓ **100%** R 0.11%	✓ **100%** R 0.23%	✓ **100%** R 0.34%	✓ **100%** R 0.46%
Colour within ΔE2000 20 of #798A9F · 6,551 match	✓ **100%** R 0.05%	✓ **100%** R 0.08%	✓ **90%** R 0.14%	✓ **87%** R 0.20%	✓ **80%** R 0.24%
Strict (all) shape AND glaze AND colour · 153 match	✓ **100%** R 1.96%	✓ **100%** R 3.27%	✓ **90%** R 5.88%	✓ **87%** R 8.50%	✓ **80%** R 10.46%
Weighted nDCG your priority	**1.00**	**1.00**	**0.98**	**0.97**	**0.95**

✓ / ✗ = Hit@K · % = Precision@K · R = Recall@K · measured on this search (evaluation never changes the order)